

4.2.3 The Initial Litterman Prior

The prior covariance matrix is built up by a sequence of modifications of an initial prior. The initial prior makes each scalar component of the $\beta^*(t; i, \cdot, \cdot)$ vector independent of all the others (i.e., it makes the covariance matrix $W[0]$ diagonal) and sets the variance according to

$$(4) \quad \sqrt{\text{Var}[\beta^*(0; i, j, s)]} = \frac{\sigma(i)}{\sigma(j)} \pi_1 \pi_2^{\delta(i,j)} \exp[-\pi_3 \log(s)],$$

$$j = 1, \dots, k + 1.$$

Here, $\sigma(i)$ is a parameter measuring the scale of fluctuations in variable i , taken in practice as the residual standard error from a univariate fifth-order VAR fit to the entire sample for $i \leq k$. For $j = k + 1$, there is only an $s = 1$ term, as the corresponding a is the “constant term” (here actually not a constant but time varying). For this term, $\sigma(j + 1) = 1/\pi_4$, another unknown parameter. The function $\delta(i, j)$ is the Kronecker delta, one for $i = j$, zero otherwise. Here, as elsewhere in this paper, the parameters π_i are “unknown constants.” In principle, we should specify a prior over them to complete a Bayesian framework for inference. However, because doing so would be inconvenient, and because we expect that our prior on them would be uninformative (i.e., we do not know much about them a priori), we integrate over these parameters informally.

4.2.4 The Dummy Initial Observation

The range of differences in observed dynamic behavior for economic time series is fairly large, and, indeed, a reasonable prior specification for the standard error of $\beta^*(0; i, i, 1)$ is about 0.16. But then this component of uncertainty about β^* alone accounts for an implied standard error of forecast for $X_i(1)$ amounting to 16 percent of the initial level of X_i . Since the random components in the other elements of $\beta^*(0)$ are all independent of this one, they all serve only to increase the implied forecast errors. We are not in fact this uncertain about the accuracy of naive random walk forecasts (which is what our initial forecasts, based on prior means for β^* , will be). We are unsure of whether our prior means are exactly right, coefficient by coefficient, but we find it much more likely that the best forecasting model will be one that implies that naive no-change forecasts will be fairly good than that it will be one that implies that great improvements on a no-change forecast are possible. If coefficients deviate from their prior means, we expect that other coefficients

where the data have in fact all been collected as unit averages) is a notably more accurate naive standard, however. Observe that the Theil U 's (for a definition, see the note to table 4.2 below) obtained by using the correct AR in place of a discrete random walk AR for a process that is actually a unit-averaged Wiener process would be, at forecast horizons 1–4, .933, .9732, .9832, and .9878.

will deviate in an offsetting way, with the result that naive no-change forecasts will still be fairly accurate.

To capture this aspect of prior beliefs, we need to introduce appropriate off-diagonal elements into $W(t; i)$ while leaving the diagonal elements relatively undisturbed. One easy way to do this is to introduce a “dummy observation” in which the prior is modified by feeding it into a Kalman filter that takes as observed data for $X(t - s)$, $s = 1, \dots, m$, the actual m initial values of X from the sample and for $X_i(t)$, not the actual $X_i(m + 1)$, but instead the model’s own forecast, based on the prior mean for β^* , of $X_i(m)$. The data in this dummy observation are weighted by a parameter π_s , which can be expected to be best taken to be near one if the variances of the $u(t; i)$ disturbances have been specified as near to the variance of forecasts from a naive random walk model. Because the Kalman filter finds that, with these artificial data, the prior mean generates perfect forecasts, the Kalman filter makes the posterior mean the prior mean. Only the variance matrix of the prior mean is changed. The change is of rank one and in practice turns out to have only modest effects on diagonal elements of W .

In most previous published work with BVARs, there has been a “sum-of-coefficients” modification to the prior. That modification can be characterized as a sequence of Kalman filtering operations indexed by $j = 1, \dots, k$, in each of which $X(t - s)$ is set to zero for $s = 1, \dots, m$, except for $X_j(t - s)$, $s = 1, \dots, m$, all of which are set to one, while $X_i(t)$, the dependent variable, is set to one if and only if $j = i$. Because most economic time series are smooth, $X(t - s)$ and $X(t - s - 1)$ have similar values. Thus, the dummy initial observation used here is approximately a linear combination of the dummy observations used in imposing the sum-of-coefficients modifications. In practice, the dummy initial observation seems to reduce or eliminate the usefulness of sum-of-coefficients dummy observations. This point is substantively important because heavily weighted sum-of-coefficients dummy observations push the model toward a limiting form written entirely in terms of differences, which eliminates all long-run relations across variables. Putting the same point another way, the old sum-of-coefficients dummy observations pulled the model toward a form with as many unit roots as variables and no cointegration, while the current dummy initial observation pulls the model toward a form with unit-root nonstationarity in all variables without down-weighting the possibility of cointegration.

This dummy-initial-observation idea was discovered in the process of adapting BVAR methodology to a context where the number of series available for a model increases at several dates scattered through a historical sample. A natural approach to such a situation is to begin with a prior for a model with all the variables that will eventually be available, padding the data for variables that are initially unavailable with zeros. Applying the Kalman filter to the padded data is equivalent to applying it to a smaller model. The

prior means and variance matrix of the coefficients on unavailable variables are left unaltered by the Kalman filter when the data for them is set at zero. However, at the time when data on a series do become available, the prior shows an exaggerated version of the problem described above as motivating the dummy initial observation. The new data multiply large prior variances on individual coefficients to imply large forecast errors, and the uncertainty about coefficients on the newly entering variable shows no correlation with uncertainty about coefficients on the variables already in the model. We know in fact that the small model estimated up to this point is a good forecasting model, and the availability of data on a new variable has not made its forecast accuracy worse. To make the prior reflect this knowledge, a dummy observation, in which the prior mean coefficients at t are presented to the Kalman filter as making perfect forecasts for $t + 1$, is appropriate. The prior mean coefficients are all zero for the newly introduced variable, so the dummy observation expresses confidence in the small model estimated without the new variable. Covariances between coefficients are created by the dummy observation, so that deviations from zero in coefficients on the new variable imply likely offsetting changes in other coefficients to leave forecasts from the previously estimated small model fairly close to those of the expanded model.

4.2.5 Relative Tightness on Durable Goods Prices

There is an *a priori* basis for expecting that prices of durable goods frequently traded in open markets will follow stochastic processes well approximated as Wiener processes over short time spans. Thus, our prior mean is inherently more attractive for such variables than, for example, for GNP or unemployment. I therefore introduce into (4) an additional multiplicative factor

$$(5) \quad \pi_j \text{IDGP}(j),$$

where $\text{IDGP}(j)$ is zero for variables that are not durable goods prices and one for variables that are. The latter are taken to be the value of the dollar, stock prices, and commodity prices. A case could be made for including three-month Treasury-bill rates and M1 in this list, but they were left out as not actually being prices of durable goods.

4.2.6 Inflation Neutrality

In theoretical models without money illusion, the price level can change without any effect on real variables. If the data show persistent changes in the price level, price-level neutrality implies certain restrictions on the coefficients of the model. In particular in a log-linear model, coefficients on the right-hand-side nominal variables should sum to one in equations with nominal dependent variables and to zero in equations with real dependent variables. The prior mean for the coefficients in this model already almost satisfies this restriction since only coefficients on own lags have nonzero means and these